

Figure 1: PhoneGap's structure

- 4) Install Java, Ant and Android Studio according to the platform on which you want to develop the app.
- 5) Update the path variables in Mac/Windows.
- 6) Test PhoneGap by entering the following commands in a terminal:

```
cd /Users/YOURUSERNAME/Desktop
phonegap create MyApp com.test.myapp TestApp
cd TestApp
phonegap local build android
```

- 7) If it is installed successfully, you will get the following results:

```
[phonegap] adding the Android platform...
[phonegap] compiling Android...
[phonegap] successfully compiled Android app
```

Environment set-up

An app contains the following items in its package:

- Configuration files
- Icons for the app
- Information or content (built using Web technologies)

Apps created using the PhoneGap build can be set up either through the Web or by setting the configuration in *config.xml*. The *config.xml* file allows us to specify the metadata about our applications.

<widget>	The widget element is the root of your XML document — it assures you that you are following the W3C specification. The following are the attributes you should set on the widget element: * id: the unique identifier for your application * version: a major/minor/patch style version * versionCode: (optional)
<name>	Name of the application.
<description>	A description for your application.

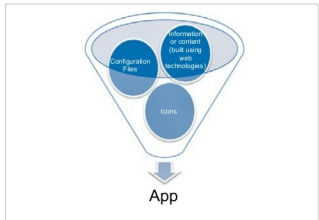


Figure 2: App structure

```
C:\install\phonegap\phonegap\phonegap-android>ruby ./droidgap "C:/
Building the JRE...
Creating Android project for target level 1
Adding new folder to project...
Generating manifest...
Copying over libraries and assets and creating phonegap.js...
Adding some application name to strings.xml...
Writing application Java code...
Complete!
C:\install\phonegap\phonegap\phonegap-android>
```

Figure 3: Command Line Output

An example is:

```
<?xml version="1.0" encoding="UTF-8" ?>
<widget xmlns = "http://www.w3.org/ns/widgets"
  xmlns:gap = "http://phonegap.com/ns/1.0"
  id = "com.phonegap.helloworld"
  versionCode = "28"
  version = "0.0.1" >
  <name>PhoneGap Hello Wold</name>
  <description>
    Hello World Example
  </description>
</widget>
```

Platforms and preferences

<platform>	Can be zero or more of platform elements. Name the platform on which we want our app to be built -- Android, iOS or Winphone.
<preference>	The preferences tag is used to personalise application configuration. It can be used to set various configuration properties like orientation, full screen, background colour, etc. These configuration properties should be in a name and value pair.

An example is:

```
<platform name="ios" />
<platform name="android" />
<platform name="winphone" />
```